

Towards Direct Tests for Consciousness in Non-Human Animals, Infants, and AI

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Abstract. The key limitation of contemporary tests for consciousness (*C*-tests) in non-human animals, infants, and AI is that they are indirect tests: they test for the presence of functions, behaviours, capacities, or states whose relation to or correlation with consciousness has to be known for implications about consciousness to follow, but which is not itself part of the test. Credence in the subsumed relation or correlation is established based on arguments, prior experiments, or theories of consciousness, with corresponding implications for the certainty and construct validity of these tests. Here, we present a method to construct direct tests for consciousness that do not depend on a known relation or correlation between consciousness and a measured property. Rather, the tests rely on the measurement of covariates of the test. Knowledge of correlations between the covariates and consciousness is not required. The method may be applicable to improve known *C*-tests. As a theoretical example, we discuss the case of infant consciousness. Our result rests on mathematical facts similar to those that have enabled the field of evaluation research in psychology.

C-tests are empirical procedures to test whether an organism or system is conscious or capable of having conscious experiences. While a range of *C*-tests are available for adult human populations, cf. e.g. (Bayne et al., 2024; Negro & Mudrik, 2026; Chalmers, 2026b), the possibility and feasibility of *C*-tests for non-canonical target populations, such as non-human animals, infants, and AI is still under intense study.

The establishment of reliable and validated *C*-tests for non-canonical target populations—and of methods to create these—is crucial for the further progress of the field. This is so because, among other things, *C*-tests:

- enable empirical progress in answering the *distribution question*—which systems or organisms are capable of having conscious experiences, and which not? Finding answers to this question matters for moral, political, and judicial reasons.
- make it possible to test *models or theories of consciousness* outside of consensus cases, and perhaps constrain or guide their development.

- may help to advance *theory-independent research programmes* in consciousness science, e.g. the search for the Neural Correlates of Consciousness (NCCs), or for the Computational Correlates of Consciousness (CCCs).¹

One feature that all previously proposed *C*-tests have in common is that they test for the presence or absence of what we will call *linking properties*—cognitive or neural functions, behaviours, capacities, or states that are related to or correlate with consciousness—rather than for consciousness itself. Any implication for the presence or absence of consciousness in an entity under investigation rests on independent knowledge about the relation or correlation between linking properties and conscious experiences. Existing *C*-tests for non-canonical target populations are, therefore, *indirect tests* for consciousness.

Knowledge about the relation or correlation between a linking property and consciousness in contemporary *C*-tests for non-canonical target populations is drawn from:

¹*C*-tests, especially of the direct kind introduced below, might be required to generalise from human NCCs or CCCs to what Chalmers (2026a) calls ‘Animal NCCs/CCCs’ that apply to the human and animal case, or ‘Universal NCCs/CCCs’ that apply universally to all entities.

1. *Arguments*, e.g. using analogical reasoning, inference to the best explanation, or a combination of both, cf. (Negro & Mudrik, 2025). Arguments may aim to establish a linking property as necessary for consciousness, sufficient for consciousness, or simply associated with consciousness.
2. *Experiments* that establish the relevance of a linking property to consciousness, or that provide information about correlations between these.
3. *Theories of consciousness* that are developed based on consensus cases, but formulated in a general enough way so as to have implications for non-consensus cases, as for example in (Butlin et al., 2023, 2025).

Dependence on these sources of knowledge places theoretical and practical limitations on contemporary C-tests:

1. *Arguments* require assumptions that may be coincidental or wrong. The fact that we can't argue our way to a full understanding of consciousness, but need scientific evidence, underwrites the limitation of arguments.
2. *Experiments* may fail to establish a rigorous enough connection to consciousness, as for example emphasized by Michel (2026)'s criticism of contemporary tests for animal consciousness.
3. *Theories of consciousness* at present are at a very initial stage, where decisive evidence for any one theory is lacking, and where the exact formulations of the theories—and with it, the implications for non-human organisms or systems—are still in flux.²

As a result, both the construct validity and the certainty of contemporary C-tests remain questionable: it ultimately remains unclear whether a C-test tests for consciousness at all, and which confidence level it gives about the presence of consciousness even if it does test for it.

Several strategies are available to mitigate this problem to some extent, e.g. cluster-based approaches (Birch, 2022; Bayne, Frohlich, Cusack, Moser, & Naci, 2023; Shea & Bayne, 2010), where a cluster of tests that make use of different linking properties with different justificatory strategies are evaluated in tandem, or statistical approaches that aim to secure novel tests for non-canonical target populations in a step-by-step fashion by control

of covariate structures or related statistical measures of association (Bayne et al., 2024). At the present stage, however, a quantification of the resulting construct validity or certainty are not available.

The ubiquity of indirect C-tests is attested to by the state-of-the-art definition, provided by (Negro & Mudrik, 2026), who define C-tests as “any protocol or methodology devised to detect specific properties that, if present, would justify higher credence in the belief that the system under test is phenomenally conscious” (p. 3). We will discuss and revise this definition in Section 10.

The goal of this paper is to present a method to construct direct C-tests, where ‘direct’ means that the tests do not depend on a known relation or correlation between consciousness and a measured property. This method does not aim at a single fixed class of tests, but is more akin to a statistical framework. This framework is applicable to many different C-tests, might provide a foundation for the development of new C-tests, and may, under the right conditions, be used to extend existing C-tests to non-canonical target populations of interest.

The presentation in this paper aims at mathematical and logical simplicity: all steps in the construction of direct C-tests presented here should be easy to follow and understandable to a wide readership. Because of the simplicity, the material here is, however, only a first step of a possible research programme aimed at developing a general methodology for the construction of direct C-tests. It is a mathematical proof-of-concept; further development may be necessary to enable progress beyond the limitations of the simple formalism presented here.

This paper is structured as follows. First, in Section 1 we explain the specific goal of the subsequent presentation, and introduce the example we will work with in that presentation, infant consciousness. In Section 2, we introduce a lean formalism for C-tests that we will make use of in what follows. Section 3 explains the main formal quantity that underlies our result. Section 4 reviews which quantities introduced up to this point are measurable, and which not. In Section 5, we explain the crux of our proposal, a formal fact similar to that of covariate adjustment in evaluation research. Section 6 explains how this formal fact can be used to estimate consciousness directly and how it makes direct C-tests possible. Section 7 summarizes how

²An example of this flux is the presentation of Global Neuronal Workspace Theory (GNWT) in (Butlin et al., 2023) in order to discuss the possibility of consciousness in AI systems. This presentation of GNWT differs significantly from earlier presentations of the theory in the scientific literature, and has different implications for the possibility of consciousness in AI systems.

this method would be applied in practice. Section 8 gives two schematic examples thereof. Section 9 contains a discussion of the limitations of the presented formalism. In Section 10, we review and discuss definitions of direct and indirect *C*-tests. Section 11, finally, offers a conclusion.

1. Context & Assumptions

Our goal in what follows is to introduce as simple a formalism as possible in order to explain—and prove—the possibility of direct *C*-tests.

To this end, we break with the mathematical tradition of aiming at the most general definition, but rather work with a single and simplified example. The goal is to use this example to explain the mathematics and the logic that make direct *C*-tests possible. Generalisation to other cases should be reasonably straightforward, and a general formalism can be constructed in subsequent work.

The example we work with is a *C*-test for infant consciousness, which for the purpose of the following explanations, we understand as follows. For details on contemporary research on infant consciousness, cf. (Passos-Ferreira, 2024; Bayne et al., 2023; Passos-Ferreira, 2023).

- **Population.** We assume that the population under consideration consists of two groups. One group—the *reference population*—comprises adult humans. The other group—the *target population*—comprises infants. The term ‘population’ here is meant in a statistical sense: it is these groups that we seek to analyse and learn about. In practice, both groups would be defined by reference to background conditions that exclude known confounders from the analysis.
- **Consciousness.** We assume that there is a fact to the matter about whether a member of the population is conscious. Here, ‘being conscious’ can be interpreted either as the *capacity for conscious experiences* in general (also called the *potential for consciousness* (Mudrik, Mylopoulos, Negro, & Schurger, 2023)), or as the *presence of conscious experiences* at the time of testing. Different tests have different conceptual targets, but from the perspective of the following formalism, there is no difference.
- **Assessing Consciousness.** We assume that there are means to assess whether members of the reference population—here, members of

the adult human populations—are conscious. These means may include introspection, reports, or behaviour, among other methods, cf. Chalmers (2026b). The fact that such means of assessment are not available for non-canonical target populations, including non-human animals, infants, and AI, is what makes *C*-tests necessary.

- **C-Test.** A *C*-test is an empirical procedure meant to answer the question of whether members of a non-canonical target population are conscious. We make two simplifying assumptions. The first assumption is that a *C*-test can, in principle, be applied to any member of the reference or target population. This requirement is necessary for the simple formalism presented in what follows to be well-defined, but is neither a practical requirement of the application of this formalism, nor necessary for generalisations thereof. The second assumption is that the test result is binary: it is either 0 or 1, indicating a positive or negative test result.³
- **Goal.** To keep the following presentation as simple as possible, we take the goal of a *C*-test to be the inference of whether members of the target population are conscious. In terms of the example of infant consciousness, this is the question of whether infants are conscious. The formalism we present in what follows gives a quantitative answer to this question: the probability with which a member of the target population is conscious. Note, however, that this target sits at the population level, not at the entity level; the formalism cannot, at its present stage of development, be applied to distinguish between individual members of the target population. Finer analyses require changes in the underlying target populations.

We would like to emphasize that none of the simplifications mentioned here is necessary. Generalizing the methodology to avoid the assumptions mentioned above should be reasonably straightforward, but a presentation of a general formalism would greatly reduce the accessibility of the paper.

A quick note to the mathematically-minded reader. To make this text as accessible as possible, we will refrain from giving any mathematical details that are not strictly necessary for the explanation of the idea to a diverse readership. For example, we will simply use the notation $p(Y|C = c)$ to denote a conditional probability distribution of a variable

³A point worth clarifying may be the difference between the applicability of a *C*-test and its conclusions. A *C*-test is always designed so that its empirical procedures—the questions or tasks that an entity has to address, or the neuroimaging scanning protocols—can be applied to some members of its target population. The crucial question is what the result of this application means in relation to consciousness. Our idealisations above concerns the applicability to *all* members of the reference and target population, and the binary nature of the result.

Y relative to a fixed value c of a variable C , without specifying any formal assumptions for this to make sense, and without specifying formal details that this distribution may need to satisfy. We will also not discuss or introduce probability spaces that would have to be defined for the mathematical explication of those variables, and will only say very little about the definition of the probability p . This text is not written for mathematicians.

2. Formalism for Direct C-Tests

In this section, we introduce a lean formalism to express our result.

2.1. Test Population. We denote by U the (statistical) population of interest. As explained above, we will work with infant C -tests as an example. Therefore, U comprises suitable choices of adult and infant subpopulations. We will denote members of the population U by u .

Furthermore, we denote by X a random variable that indicates the population subgroup to which a member of the population belongs. For a member u of the population, $X(u) = 1$ means that the member is an adult, whereas $X(u) = 0$ means that the member is an infant.

2.2. Consciousness. We denote by C a random variable that indicates consciousness. $C(u) = 1$ means that member u of the population is conscious, while $C(u) = 0$ means that member u is not conscious. It is this variable that C -tests are intended to test for.

We denote by π what could be called the *prevalence for consciousness*. Specifically, we denote by

$$\pi_1 = p(C = 1 | X = 1)$$

the prevalence for consciousness in the adult population. It is the probability with which any adult human has the capacity for consciousness. Analogously, we denote by

$$\pi_0 = p(C = 1 | X = 0)$$

the prevalence for consciousness in the infant population. It is the probability with which an infant has the capacity for consciousness. The latter, π_0 is what we would like to estimate or infer empirically if we use C -tests with a population-level goal.

2.3. C-Test. Next, we introduce statistics that describe a C -test.

First, we denote by Y a C -test. Because we assume that this test can be applied to any member of the population, we can take Y to be a random variable. Because we have assumed that C -tests are binary, so is Y . We take $Y(u) = 1$ to indicate a positive test result for the member u of the

population, while $Y(u) = 0$ indicates a negative test result. The point of the following method (and equally so of methods based on linking properties) is to establish what these results *mean*.

Where statistics is concerned, this meaning is given by the test's sensitivity and specificity. We distinguish these for the infant and adult subpopulations. For the adult subpopulation, we denote the test's sensitivity by se_1 , and the test's specificity by sp_1 . They are given by:

$$se_1 = p(Y = 1 | C = 1, X = 1)$$

$$sp_1 = p(Y = 0 | C = 0, X = 1).$$

For the infant case, we denote the test's sensitivity by se_0 , and the test's specificity by sp_0 . They are given by:

$$se_0 = p(Y = 1 | C = 1, X = 0)$$

$$sp_0 = p(Y = 0 | C = 0, X = 0).$$

2.4. Covariates. For the method presented below, it is important to keep track of the covariates of the test, where a covariate is "a variable (= a quantity that can change) that may affect the result of what is being studied" (Cambridge Dictionary, 2026, entry on 'covariate'). A covariate of a C -test Y , therefore, is a quantity that can change the result of the C -test.

We will denote the covariates of Y by $Z_1, \dots, Z_n$, and summarize them in terms of a multi-valued random variable Z that contains *all* covariates of a C -test under consideration, other than consciousness. This variable is $Z = (Z_1, \dots, Z_n)$. We denote a specific value of Z by z , so that we have $z = (z_1, \dots, z_n)$, where each z_i is a specific value of Z_i .

We will call all members of the population whose covariates Z have a single value z a *stratum* of the population. All members in a stratum have the same covariates of the test under consideration.

For reasons explained in Section 5, it will be crucial to keep track of covariates in the sensitivity and specificity of the test. We therefore introduce covariate-specific sensitivities and specificities as

$$se_1(z) = p(Y = 1 | C = 1, X = 1, Z = z)$$

$$sp_1(z) = p(Y = 0 | C = 0, X = 1, Z = z)$$

for the adult case, and as

$$se_0(z) = p(Y = 1 | C = 1, X = 0, Z = z)$$

$$sp_0(z) = p(Y = 0 | C = 0, X = 0, Z = z)$$

for the infant case.

3. Change-of-Population Effect

The decisive ingredient in the construction of direct C-tests in what follows is the measurement of which effect a change of population has on the C-test. We call this the *change-of-population effect*, and denote it by CPE. It is defined as

$$\text{CPE} = E(Y|X = 1) - E(Y|X = 0), \quad (3.1)$$

where E denotes the expectation value. The first term, $E(Y|X = 1)$, is the conditional expectation value of the test in the adult population. The second term, $E(Y|X = 0)$, is the conditional expectation value of the test in the infant population.

The definition of the CPE is inspired by the so-called average treatment effect (ATE) in the statistical field known as evaluation research (Döring, 2023; Stockmann & Meyer, 2013; Rossi, Lipsey, & Freeman, 2003), which quantifies the average effect that a treatment has on a patient characteristic of interest.

Using only what we have defined above, the change-of-population effect can be expressed in terms of the sensitivities, specificities, and prevalences as follows. We can expand the conditional expectation value in the adult population as

$$\begin{aligned} E(Y|X = 1) &= 1 \cdot p(Y = 1|X = 1) + 0 \cdot p(Y = 0|X = 1) \\ &= p(Y = 1|X = 1) \\ &= p(Y = 1|C = 1, X = 1) \cdot p(C = 1|X = 1) \\ &\quad + p(Y = 1|C = 0, X = 1) \cdot p(C = 0|X = 1), \end{aligned}$$

where in the last step we have used the law of total probability. The first term of the result of this calculation is equal to the adult sensitivity multiplied with the adult prevalence,

$$\begin{aligned} p(Y = 1|C = 1, X = 1) \cdot p(C = 1|X = 1) \\ = se_1 \cdot \pi_1. \end{aligned}$$

The second term can be expanded in terms of the adult sensitivity and adult prevalence as

$$\begin{aligned} p(Y = 1|C = 0, X = 1) \cdot p(C = 0|X = 1) \\ = (1 - p(Y = 0|C = 0, X = 1)) \\ \cdot (1 - p(C = 1|X = 1)) \\ = (1 - sp_1) \cdot (1 - \pi_1), \end{aligned}$$

where sp_1 is the adult specificity and π_1 is, again, the adult prevalence. In total, this gives

$$E(Y|X = 1) = se_1 \cdot \pi_1 + (1 - sp_1) \cdot (1 - \pi_1).$$

The exact same calculation for the infant case (replacing $X = 1$ by $X = 0$ in the above) gives

$$E(Y|X = 0) = se_0 \cdot \pi_0 + (1 - sp_0) \cdot (1 - \pi_0).$$

In sum, the change-of-population effect is given by

$$\begin{aligned} \text{CPE} &= se_1 \cdot \pi_1 + (1 - sp_1) \cdot (1 - \pi_1) \\ &\quad - se_0 \cdot \pi_0 - (1 - sp_0) \cdot (1 - \pi_0) \end{aligned} \quad (3.2)$$

4. Measurable and Unmeasurable Quantities

Before moving on with the analysis, we need to discuss which of the above quantities can, in principle, be measured or estimated, and which not.

A C-Test Y can be applied to all members of the population, which is why its expectation values across sub-populations, $E(Y|X = 1)$ and $E(Y|X = 0)$, and therefore also the change-of-population effect CPE as defined in (3.1), can be measured.

Furthermore, because consciousness can be assessed in the adult population, the adult prevalence π_1 , the adult sensitivity se_1 , and the adult specificity sp_1 can all be measured. (In practice, it may be the case that for a suitable chosen adult population, π_1 is close to 1.) The same holds true, at least in principle, for the covariate-specific adult specificity and adult sensitivity.

Because consciousness in infants cannot be independently assessed, the infant sensitivity and infant specificity of the test, se_0 and sp_0 , cannot be measured. The same holds true for the target quantity of the test, the infant prevalence π_0 .

5. What Makes Direct C-Tests Possible

In the last section, we have explained that the infant sensitivity and infant specificity of a C-test, se_0 and sp_0 , cannot be measured. And if se_0 and sp_0 cannot be measured, one might think, it is impossible to use the test results to make inferences about the target quantity, the infant prevalence of consciousness, π_0 .

Due to a statistical method otherwise known as covariate adjustment (Rubin, 1974), however, it is possible to obtain knowledge about the infant specificity and infant sensitivity of the test even though they cannot themselves be measured. We now explain how.⁴

As explained in Section 2.4, a covariate is “a variable (= a quantity that can change) that may affect the result of what is being studied” (Cambridge Dictionary, 2026, entry on ‘covariate’). We have chosen to denote the covariates of the test Y by

⁴Our reasoning is, in fact, simpler than the usual means of covariate adjustment. This is the case because covariate adjustment, as usually applied, utilizes the fact that if all covariates of a test are controlled for, the assignment to a treatment variable in a covariate strata is, de facto, random. In our case, the ‘treatment variable’ is replaced by a population variable, which is descriptive and not causally efficacious. This makes our argument even simpler than in the usual case. Cf. Section 9 for more on that.

$Z = (Z_1, \dots, Z_n)$. Because Z contains all covariates, the test Y only depends on Z and consciousness. If we denote this dependence by f —a function or probability channel—we can express this as

$$Y = f(C, Z_1, \dots, Z_n), \quad (5.1)$$

or more directly as

$$y = f(c, z_1, \dots, z_n),$$

where c is the value of the consciousness variable, z_i are values of the covariates, and y indicates a test result.

These formulae express the fact that the result of the test depends only on whether a subject is capable of being conscious, and on the specific covariate stratum that the subject is part of, which is a consequence of the inclusion of *all* covariates in these formulae. Because we have included all covariates, no further details matter. If there is residual (“true”) randomness, f is a probability channel. If not, f is a function.

We can now introduce the population variable X that denotes whether a member u is part of the adult or infant population. Because we have already included all covariates above, conditioning on any additional variables leaves the result of the test fixed. There is, simply put, nothing to vary once all covariates are fixed. Therefore, we have

$$\begin{aligned} p(Y = y|C = c, X = 1, Z_1 = z_1, \dots, Z_n = z_n) \\ = p(Y = y|C = c, X = 0, Z_1 = z_1, \dots, Z_n = z_n) \end{aligned}$$

for all c , all y , and all z_i .

Switching back to the variable Z introduced above, $Z = (Z_1, \dots, Z_n)$ and $z = (z_1, \dots, z_n)$, and setting $y = 1$ and $c = 1$, the last formula implies

$$\begin{aligned} p(Y = 1|C = 1, X = 1, Z = z) \\ = p(Y = 1|C = 1, X = 0, Z = z), \end{aligned}$$

which if expressed in terms of the infant and adult sensitivities is

$$se_1(z) = se_0(z). \quad (5.2)$$

Similarly, setting $y = 0$ and $c = 0$ implies that

$$\begin{aligned} p(Y = 0|C = 0, X = 1, Z = z) \\ = p(Y = 0|C = 0, X = 0, Z = z), \end{aligned}$$

which in terms of the infant and adult specificities is

$$sp_1(z) = sp_0(z). \quad (5.3)$$

The upshot of this result is that if all covariates of a test are controlled for—meaning that they are kept constant—then for every specific value z of the covariates Z , $se_0(z)$ and $sp_0(z)$ known, because they are identical to $se_1(z)$ and $sp_1(z)$, respectively.

This is a logical fact that follows from the statistics alone. While it is unreasonable to assume that the sensitivity and specificity of the test are the same across the whole infant and adult populations, it is a logical truth that they are the same if *all* covariates of the tests are controlled for. Knowledge of the correlation between covariates and consciousness is not required to draw this conclusion.

As we will see next, this fact suffices to calculate the infant prevalence π_0 based on measurable quantities alone.

One limitation of the applicability of this result is the case where covariates of the test are perfect predictors of the population variable, meaning that for some value z_i of a covariate, there are only infant or only adult members of the population. We discuss this limitation and possible mitigation strategies in Section 9.

6. Estimating Consciousness

In this section, we explain how the infant prevalence, π_0 , can be calculated based on the statistical fact explained in the last section. The starting point of this calculation is Equation (3.2).

We cannot use Equation (3.2) to calculate π_0 directly because it is just a single equation with three unknowns (π_0 , se_0 , sp_0). The way forward is to consider the covariate-specific case—the case of specific values z of the covariates. As explained in the last section, in the covariate-specific case, $se_0(z)$ and $sp_0(z)$ are in fact known.

To this end, we define the covariate-specific change-of-population effect $CPE(z)$ as

$$\begin{aligned} CPE(z) = E(Y|X = 1, Z = z) \\ - E(Y|X = 0, Z = z), \end{aligned}$$

and the covariate-specific prevalences as

$$\begin{aligned} \pi_1(z) &= p(C = 1|X = 1, Z = z) \\ \pi_0(z) &= p(C = 1|X = 0, Z = z). \end{aligned}$$

Running through the exact same calculations as in Section 3, we obtain

$$\begin{aligned} CPE(z) = & se_1(z) \cdot \pi_1(z) + (1 - sp_1(z)) \cdot (1 - \pi_1(z)) \\ & - se_0(z) \cdot \pi_0(z) - (1 - sp_0(z)) \cdot (1 - \pi_0(z)). \end{aligned}$$

Making use of Equations (5.2) and (5.3), this expression simplifies to

$$\begin{aligned} CPE(z) = & (1 - se_1(z) - sp_1(z)) \cdot \pi_0(z) \\ & - (1 - se_1(z) - sp_1(z)) \cdot \pi_1(z). \end{aligned}$$

We can solve this equation for $\pi_0(z)$. This gives

$$\begin{aligned}\pi_0(z) &= \\ &= \frac{\text{CPE}(z) + (1 - \text{se}_1(z) - \text{sp}_1(z)) \cdot \pi_1(z)}{(1 - \text{se}_1(z) - \text{sp}_1(z))}.\end{aligned}\quad (6.1)$$

This equation determines the covariate-specific infant prevalence, $\pi_0(z)$, based on measurable quantities alone, and therefore makes $\pi_0(z)$ measurable.

For the target quantity π_0 , we can use the law of total probability to obtain

$$\begin{aligned}\pi_0 &= p(C = 1|X = 0) \\ &= \sum_{z \in Z} p(C = 1|X = 0, Z = z) \cdot p(Z = z|X = 0) \\ &= \sum_{z \in Z} \pi_0(z) \cdot p(Z = z|X = 0),\end{aligned}$$

where $p(Z = z|X = 0)$ is the relative frequency of stratum z in the infant population. Overall, we therefore obtain

$$\pi_0 = \sum_{z \in Z} p(Z = z|X = 0) \cdot \pi_0(z). \quad (6.2)$$

We have therefore obtained a way to calculate the target quantity—the prevalence of consciousness in the infant population, π_0 —based on measurable quantities alone.

In practice, the measurable quantities of interest— $\text{se}_1(z)$, $\text{sp}_1(z)$, and $\pi_1(z)$ —may also be estimated; precise values are not needed as long as error rates are under control. Equations (6.1) and (6.2) therefore lay the path for direct tests for consciousness in the infant population.

7. How to Apply in Practice

The results presented above are mathematical in nature. The goal of this section is to explain how these mathematical results can be applied in practice.

To keep the explanation as simple as possible, we provide a step-by-step “recipe” for the application; in practice, however, the various steps likely interdepend. Furthermore, we continue to frame the explanation in terms of infant consciousness, even though the results and procedure can be applied to other tests/target populations as well.

An application of the above result comprises the following steps:

- 1.) **Selection/Design of C-Test.** The first step is to pick an existing C -test, or alternatively design a new C -test tailored to the further steps below.

- 2.) **Determination of Covariates.** The second step is the identification of the covariates of the chosen C -test. As in other disciplines, this is a multifaceted process that involves both theoretical and empirical considerations. The result of this step is the variable $Z = (Z_1, \dots, Z_n)$ in the above formalism. For limitations that result from imperfect choices, cf. Section 9.

- 3.) **Reference Group Measurements.** The third step consists of the measurement of covariate-specific test-properties in the reference group, namely the adult sensitivity $\text{se}_1(z)$, the adult specificity $\text{sp}_1(z)$ and the adult prevalence $\pi_1(z)$, where z are the covariate strata—the individual values of the covariates.

- 4.) **Measurement of Change-of-Population Effect.** The fourth and final step consists of the measurement of the covariate-specific change-of-population effect $\text{CPE}(z)$, which is the difference between the covariate-specific expectation value of the test in the adult and infant populations.

Once these steps have been carried out, the infant prevalence π_0 —the probability that an infant is conscious—can be calculated based on Equations (6.1) and (6.2).

8. Examples

In this section, we provide two *schematic* examples of how the method we have propose here can be applied to existing tests for infant consciousness. Our goal in doing so is purely pedagogical, full-fledged applications will have to be developed with groups that actively employ tests. The discussions of the following examples are based on (Passos-Ferreira, 2024) and (Bayne et al., 2023), to which we refer for further details.

- 8.1. **Detection of Global Oddballs.** A primary source of evidence for consciousness in newborn infants is the so-called local-global paradigm (Bekinschtein et al., 2009). In a nutshell, in this paradigm, a newborn infant is presented with auditory clues that exhibit a predictable local pattern, typically “aaaaB, aaaaB, aaaaB” (Passos-Ferreira, 2024, p.1520), followed by an unpredictable “global” change in the local pattern, e.g. “aaaaA”, called a global oddball. In awake adults, detection of the global oddball causes a positive cortical response roughly 300ms after the stimulus presentation, called the P300 complex. If newborn infants exhibit a similar neural response when presented with a global oddball, this is taken as evidence that they detect the global oddball and

therefore are conscious. Indicators of global oddball detection are indeed found in 3-month old infants (Basirat, Dehaene, & Dehaene-Lambertz, 2014), newborns (Moser et al., 2020), and fetuses past 34 weeks of gestational age (Moser et al., 2021; Bayne et al., 2023).

Why does detection of a global oddball constitute evidence for consciousness in newborn infants? At the present stage of development, two types of reasons can be given:

G1.) Indicators of Correlation. While awake and attentive adults do show a P300 component when presented with a global oddball, comatose patients, sleeping adults, or inattentive adults do not (Passos-Ferreira, 2024). Furthermore, the P300 response to global oddball presentations can still be found in patients in the minimally conscious state, but cannot be found in patients with the unresponsive wakefulness syndrome (Bekinschtein et al., 2009; Bayne et al., 2023).

These findings can be interpreted as evidence that detection of a global oddball correlates with—or is predictive of—consciousness in the reference group of adult humans. If one furthermore assumes that this correlation also holds for the target population, detection of a global oddball can speak in favour of consciousness in newborn infants.

G2.) Argument that Test Target Requires Consciousness. A second strategy is to take the detection of a global oddball to test for a cognitive capacity: the ability to maintain perceptual representations over time (Passos-Ferreira, 2024), and to argue that this ability requires consciousness.

Both of these reasons turn evidence for global oddball detection in infants into evidence for consciousness in infants. But this implication is, at present, still comparably weak. There is evidence that a global oddball can be obtained in the absence of consciousness (Tzovara, Simonin, Oddo, Rossetti, & De Lucia, 2015), which puts pressure on the above-mentioned indicators of a correlation between consciousness and global oddball detection. Furthermore, the assumption that a correlation found in the reference group also holds for the target group is not empirically testable. Similarly, the argument that maintaining perceptual representations over time requires consciousness can be questioned. Because, in addition, the parameters of the P300 response in infants differ from those of the adult case (Bayne et al., 2023), the detection of global oddballs can, at present, only be “regarded

as having better than chance accuracy for detecting consciousness” (Bayne et al., 2023, p. 1142), and is seen as supporting a “best guess” (ibid.) about consciousness in infants. Further progress towards more rigorous tests is required.

The method we have presented here allows for an alternative approach to test and possibly harden the implications of global oddball evidence:

G3.) Method Presented Here. The application of the method we have presented here (covariate adjustment) requires the determination of *covariates* of the ability to detect global oddballs (viz., the P300 response to a global oddball presentation). Intuitively speaking, these are modulators of the ability to detect global oddballs. Examples of such covariates include the overall arousal level of the subjects, measures of task engagement, the capacity or use of working-memory relevant to the task, and subject characteristics. For each value of these covariates, the sensitivity and specificity of the global oddball test in the adult population, as well as the change-of-population effect, have to be measured or estimated (Steps 3.) and 4.) above). The method then allows for the calculation of the prevalence of consciousness in the target population—here, newborn infants.

While this method comes with its own challenges—mainly, in the determination of covariates (cf. Section 9)—it allows for a statistically rigorous and quantitative use of the global oddball paradigm.

8.2. Recurrent Processing. A different line of evidence regarding infant consciousness, but with opposing implications, comes from tasks that test for the presence of recurrent processing in infants. Examples for such tasks are visual backward masking (Breitmeyer & Ogmen, 2006) or object substitution masking (Enns & Di Lollo, 1997). Studies that make use of these tasks to test for recurrent processing in infants and fail to do so, such as (Nakashima, Kanazawa, & Yamaguchi, 2021), are sometimes taken as evidence against infant consciousness (Passos-Ferreira, 2024), but cf. (Fahrenfort, Scholte, & Lamme, 2007).

Why is recurrent processing taken to be a suitable test target for inference of infant consciousness? In this case, the justification is mainly theory-based:

R1.) Theories of Consciousness. Recurrent processing in the visual cortex is a necessary

condition for conscious perception according to Recurrent Processing Theory (RPT, also known as Local Re-Entry Theory) (Lamme, 2006).

Conclusions regarding infant consciousness in this case therefore rely on the truth of RPT, which to date is not more or less likely than the truth of other scientific theories of consciousness, cf. (Seth & Bayne, 2022) for an overview. The method presented here may provide a path towards theory-independent assessment of the implications of the visual backwards masking or object substitution masking experiments:

R2.) Method Presented Here. As in the case of global oddball detection (Section 8.1), the application of the method we have presented here to visual backward or object substitution masking requires the determination of *covariates* of these tasks. For each value of these covariates, the sensitivity and specificity of the task in a suitable adult reference population, as well as the change-of-population effect, have to be measured or estimated (Steps 3.) and 4.) in Section 7).

8.3. Conclusions. For both the global oddball and recurrent processing tests, rigorous and cogent *applications* of the tests exist, but justificatory strategies regarding the implications for infant consciousness are still weak. They depend on limited and partially contested empirical evidence (G1.), defeasible arguments (G2.), and theories of consciousness whose truth is not established (R1.). As a consequence the construct validity and certainty of the tests (whether they test for consciousness and which confidence level they give regarding the presence or absence of consciousness in infants) can still be questioned.

The method we have presented here is meant to *expand* the repertoire of justificatory strategies of these tests. It offers a complementary way to work with these tests, which requires additional measurements (of the covariate-specific test quantities), but circumvents dependence of the interpretation of the tests' results on empirical correlations, arguments, or theories of consciousness.

The largest draw-back of the method we have presented is the determination of covariates of the test. In the next section, we address this and other limitations.

9. Limitations

In this section, we address the major limitations of the result we have obtained in the previous sections.

The first limitation derives from the nature of the population variable. In the derivation presented in Sections 3 to 6, we have assumed that the population variable X that distinguishes between the infant and adult populations is a *descriptive* variable. As a consequence, X does not appear in Equation (5.1). When expressed in causal terms, this means that the population itself to which a subject belongs does not have a causal effect on the test result; only specific properties that differ between the populations do.

This limitation means that the mathematics we have presented above does not apply to the case where membership to the population itself—over and above physiological, neural, cognitive, or computational differences between the populations—is a causal factor that determines the result of the test. A more general formalism would have to be developed to treat this case as well.

Such a formalism can likely be developed. This is so because the field of evaluation research that provides the inspiration for the proposal developed here does, in fact, already contain statistical methods to treat exactly this case: the treatment variable in evaluation research, which corresponds to the population variable in our case, is (of course) a causal factor that determines the test. More advanced covariate adjustment methods are available to address exactly this case. We do not make use of them here because the rationale in this paper is to present the simplest possible case.

The second limitation derives from the assumption that, logically speaking, *all* covariates of a test have to be measured for the above result to hold. This limitation is of greater relevance, as it is unclear whether this can be achieved in practice. Furthermore, unlike for the first assumption, there is no mathematical fix in the field of evaluation research that can be applied. Methods there, too, rest on the formal assumption that all covariates of a test are controlled for.

Does this mean that the procedure proposed above is not applicable in practice? Arguably not. As in the case of evaluation research, controlling for the known covariates is likely enough to get reliable results. If there is doubt about further covariates the results obtained in practice might not live up to the maximum standards or rigour, but if all known covariates are controlled for, they will still live up to the standards of rigour that are applied in the fields of medicine or psychology in the drug or therapy approval processes.

A third limitation concerns covariates that are perfect predictors of the population variable, meaning that some values of the covariates are possessed

exclusively by infant or exclusively by adult members of the population. This poses a problem because the measurement of the covariate-specific change-of-population effect, $CPE(z)$, requires the measurement of the conditional expectation value of the test in both the infant and adult population, for all covariate strata.

While strong asymmetries in the relative size of covariate strata across both populations can be handled by statistical tools such as propensity score matching (Rosenbaum & Rubin, 1983, 1985), if a *C*-test has covariates that perfectly predict the population subgroup, it is likely that covariate-adjustment methods cannot be applied to that test. If a test with this property is meaningful at all, it is likely that it has to be modified for direct testing to be applied.

A final limitation that we have mentioned before is that the method in its present form can only be used for population-level testing; application to individuals within a population is not possible with the formalism presented here.

Most if not all of these limitations can likely be resolved, or at least mitigated, with a more powerful statistical formalism.

10. Questions of Definition

The state-of-the-art definition of *C*-Tests is given by Negro and Mudrik (2026), according to whom a *C*-Test is “defined as any protocol or methodology devised to detect specific properties that, if present, would justify higher credence in the belief that the system under test is phenomenally conscious” (p. 3). The “specific properties” referred to here are what we have called linking properties above; they justify higher credence in the belief that a system is phenomenally conscious because of arguments, experiments, or theories that establish a connection or correlation between a property and conscious experience. Negro and Mudrik (2026)’s definition, therefore, is a definition of indirect *C*-tests, and does not cover the type of *C*-test we aim at here.

Because explicit definitions are helpful both for pedagogical reasons and further progress, in this section, we briefly propose an updated definition of *C*-tests, as well as a distinction between direct and indirect *C*-Tests that is inspired by (though not equal to) (Wiese, 2026).

As a first step, we propose to make use of the antecedently understood notion of tests in conceptualizing *C*-tests, so as to help to (ultimately) draw a distinction between markers (Passos-Ferreira, 2024;

Birch, 2022, 2024; Birch, Burn, Schnell, Browning, & Crump, 2021; Fink, 2024; Brown, Paul, & Birch, 2024; Bayne et al., 2023), indicators (Wiese, 2026; Irvine, 2020; Andrews, Birch, & Sebo, 2025), and tests (Elamrani & Yampolskiy, 2019; Dung, 2022; Birch, Schnell, & Clayton, 2020) for consciousness in non-canonical target populations. Generalizing Negro and Mudrik (2026)’s notion of “protocol or methodology” (p. 3) to “experimental procedure”, and following moral philosophy in using the term “entity” to denote an organism or system, we arrive at the following definition.

Def 1. A **C-test** is an experimental procedure that tests whether an entity (=organism or system) is phenomenally conscious.

Following this lead, we can take Negro and Mudrik (2026) to provide a definition of a subclass of *C*-tests, namely *indirect C*-Tests, and adapt their definition to the above as follows.

Def 2. An **indirect C-Test** is a *C*-test that tests for specific properties that justify higher credence in the belief that the entity under test is phenomenally conscious in virtue of independent knowledge or independent evidence about the relation or correlation between these properties and consciousness.

A direct *C*-test could then be understood as the complement of indirect *C*-Tests.

Def 3. A **direct C-Test** is a *C*-test that is not indirect.

This definition of direct and indirect *C*-tests sits nicely with the intuition behind (Wiese, 2026)’s definitions of direct and indirect indicators of consciousness in AI systems. The essential difference between direct and indirect indicators of consciousness, according to (Wiese, 2026)’s definitions, is that evidence for indirect indicators does not, in and by itself, increase the “degree of confirmation” (p. 9) for the hypothesis that an AI system is conscious; it only does so in conjunction with evidence for other indicators. Evidence for direct indicators, in contrast, increases the degree of confirmation by itself; having evidence for other indicators as well does not increase the degree of confirmation from that indicator.

This difference aligns with the distinction above because tests for linking properties do not, in and by themselves, increase the degree of confirmation that an entity is conscious, but they do so if further information or further evidence about the relation between a linking property and consciousness is available. *C*-tests as developed here, in contrast,

⁵It is likely the case that Wiese (2026)’s formalism is not meant to include background knowledge as provided by arguments or theories of consciousness; the probability-rising feature is only defined relative to evidence for other indicators. To explicate the

do increase the degree of confirmation in and by themselves, independently of further information or evidence from other sources.⁵

11. Conclusion

Contemporary tests for consciousness (*C*-tests) in non-human animals, infants, and AI are indirect tests: they test for the presence of *linking properties*—functions, behaviours, capacities, or states—whose relation to or correlation with consciousness has to be known; implications for the presence or absence of consciousness in these target populations depend on independent experiments, theories, or arguments that establish these relations or correlations.

The goal of this paper was to show that another class of *C*-tests is possible as well: direct *C*-tests that do not rely on relations or correlations between a tested property and consciousness. The method presented here, which may only be one among many similar ones, instead makes use of a statistical procedure that has proven productive in testing for the efficacy of treatments in the field of evaluation research, known as *covariate adjustment*.

The crucial point about the method presented here is that no knowledge about the correlation or connection between the test and consciousness in the target population (e.g. infants, non-human animals, AI) is required. It is sufficient to measure the test sensitivity and specificity in a reference population (usually, adult humans) and covariates of the test. The mathematics we presented here then give a formula to calculate the probability that members of the target population are conscious.

The results presented in this paper are meant to be a first step only. They show that direct *C*-tests are possible, and how they can be constructed, but are aimed at optimizing readability and simplicity, rather than generality. A more general—and potentially more powerful—framework may have to be constructed in future work.

AI Declaration. No AI was used in writing this paper. While I endorse the use of AI for literature review and understanding of complex papers, I believe that the essential records of our scientific work should reflect an author's own means of expressions and ways of thinking.

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